

ANSWER BOOK

I AGREE TO ABIDE BY THE SENATE POLICY ON CONDUCT OF ASSIGNMENTS, MID-TERM EXAM, AND FINAL EXAM

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Assignment on **Fuzzy Sets**:

Q2a. Solution:

$$\begin{array}{l|l} X_1 = [1, 2, 3] & A_1 = 0.2 | 1 + 0.5 | 2 + 0.7 | 3 \\ X_2 = [3, 4, 5, 6] & A_2 = 0.2 | 3 + 0.5 | 4 + 0.7 | 5 + 0.2 | 6 \\ \\ y = f(x_1, x_2) & \\ Y = [4, 5, 6, 7, 8, 9] & B = f(A_1, A_2) = ? \end{array}$$

The calculations for the two-dimensional to one-dimensional universal mapping is shown below:

k	x_1	$\mu_{X_1}(x_1)$	x_2	$\mu_{X_2}(x_2)$	y	$\min[\mu_{X_1}(x_1), \mu_{X_2}(x_2)]$	$\mu_B(y)$
1	1	0.2	3	0.2	4	0.2	0.2
2	1	0.2	4	0.5	5	0.2	0.2
3	1	0.2	5	0.7	6	0.2	0.5
4	1	0.2	6	0.2	7	0.2	0.5
5	2	0.5	3	0.2	5	0.2	0.2
6	2	0.5	4	0.5	6	0.5	0.5
7	2	0.5	5	0.7	7	0.5	0.5
8	2	0.5	6	0.2	8	0.2	0.7
9	3	0.7	3	0.2	6	0.2	0.5
10	3	0.7	4	0.5	7	0.5	0.5
11	3	0.7	5	0.7	8	0.7	0.7
12	3	0.7	6	0.2	9	0.2	0.2

Since there are two input membership functions, $\mu_{X_1}(x_1)$ and $\mu_{X_2}(x_2)$ their minimum membership values are taken as the input membership value.

From the above calculations, it is clear that there are ordered pairs given by:

- (a) $[x_1, x_2] = [1, 4]$ with membership function $[\mu_{X_1}(x_1), \mu_{X_2}(x_2)] = [0.2, 0.5]$ and $[x_1, x_2] = [2, 3]$ with membership function $[\mu_{X_1}(x_1), \mu_{X_2}(x_2)] = [0.5, 0.2]$ which are mapped to the same point $y = 5$.
- (b) $[x_1, x_2] = [1, 5]$ with membership function $[\mu_{X_1}(x_1), \mu_{X_2}(x_2)] = [0.2, 0.7]$, $[x_1, x_2] = [2, 4]$ with membership function $[\mu_{X_1}(x_1), \mu_{X_2}(x_2)] = [0.5, 0.5]$ and, $[x_1, x_2] = [3, 3]$ with membership function $[\mu_{X_1}(x_1), \mu_{X_2}(x_2)] = [0.7, 0.2]$ which are mapped to the same point $y = 6$.
- (c) $[x_1, x_2] = [1, 6]$ with membership function $[\mu_{X_1}(x_1), \mu_{X_2}(x_2)] = [0.2, 0.2]$, $[x_1, x_2] = [2, 5]$ with membership function $[\mu_{X_1}(x_1), \mu_{X_2}(x_2)] = [0.5, 0.7]$ and, $[x_1, x_2] = [3, 4]$ with membership function $[\mu_{X_1}(x_1), \mu_{X_2}(x_2)] = [0.7, 0.5]$ which are mapped to the same point $y = 7$.
- (d) $[x_1, x_2] = [2, 6]$ with membership function $[\mu_{X_1}(x_1), \mu_{X_2}(x_2)] = [0.5, 0.2]$ and $[x_1, x_2] = [3, 5]$ with membership function $[\mu_{X_1}(x_1), \mu_{X_2}(x_2)] = [0.7, 0.7]$ which are mapped to the same point $y = 8$.

Thus, their respective maximum membership values must be taken. The fuzzy set B can be obtained from the extension principle as

$$B = 0.2 \mid 4 + 0.2 \mid 5 + 0.5 \mid 6 + 0.5 \mid 7 + 0.7 \mid 8 + 0.2 \mid 9$$

Thus, the sum of all the membership function values of the derived fuzzy set B is

$$\begin{aligned} \text{sum} &= 0.2 + 0.2 + 0.5 + 0.5 + 0.7 + 0.2 \\ \text{sum} &= 2.3000 \end{aligned}$$

Q2b. Solution:

Based on your calculations in Q2a we can see that,

- The number of one-to-one mappings is two: for point $y = 4, 9$.
- The number of two-to-one mappings is two: for point $y = 5, 8$.
- The number of three-to-one mappings is two: for point $y = 6, 7$.

Therefore, all the statements are correct.